

DURE-ST: DUAL-RELATION SPECTRO-TEMPORAL MODELING FOR SPEECH DEEFAKE DETECTION

Shaole Li, Siqing Qin, Youzhi Tu, and Kong Aik Lee[†]

Department of Electrical and Electronic Engineering
The Hong Kong Polytechnic University, Hong Kong SAR

ABSTRACT

Previous speech deepfake detectors can adaptively capture spectro-temporal dependencies through graph attention, yet they largely overlook the co-variation between spectral and temporal representations. To address this gap, we construct a normalized affinity graph from their joint covariance and apply polynomial graph filtering to capture higher-order covariance-induced dependencies. We first develop Cov-ST to isolate the contribution of covariance-based relational modeling. Although it improves detection performance, its sensitivity to the polynomial order suggests limited robustness when covariance relations are modeled alone. We therefore propose DuRe-ST, which jointly exploits covariance-induced and graph-attention-induced relations to capture complementary second-order co-variation and adaptive spectro-temporal dependencies. Experiments show that DuRe-ST achieves an average relative EER reduction of 25.9% over XLSR-AASIST on the ASVspoof benchmarks and 28.4% across four cross-dataset benchmarks with only 4–8k additional trainable back-end parameters. It further outperforms the strongest publicly available comparison models by 2.2–13.2% in relative EER on four benchmarks, while remaining smaller than the publicly available models considered.

Index Terms— speech deepfake detection, spectro-temporal modeling, covariance, graph attention

1. INTRODUCTION

Deepfake speech generated by text-to-speech (TTS) and voice conversion (VC) methods raises concerns about voice impersonation, fraud, and misinformation [1, 2]. These risks have driven advances in speech deepfake detection (SDD), including new model architectures [3, 4], training strategies [5, 6], and data augmentation methods [7]. Many recent detectors use pretrained self-supervised speech models, such as XLS-R [8] and WavLM [9], as front-end feature extractors and rely on a task-specific back-end to model spoofing cues.

These back-ends model spectral and temporal information in different ways. Convolutional models, such as CG-Res2Net [10] and Nes2Net [11], capture local acoustic patterns through multi-scale feature extraction. Transformer- and Conformer-based models use self-attention to model long-range context [4, 12], while Mamba-based detectors employ selective state-space models for efficient sequence modeling [13]. AASIST takes a different approach by organizing spectral and temporal features as heterogeneous graph nodes and using graph attention to model relations within and across the two domains [3, 14]. As a mainstream backbone, this heterogeneous modeling makes AASIST natural for studying how spectral and temporal features interact in SDD.

However, AASIST models spectro-temporal relations mainly through learned attention weights. These weights capture the relevance between nodes but do not explicitly represent covariance, a second-order statistic that describes how spectral and temporal features vary together. This omission matters because synthetic speech may contain inconsistent or unnaturally over-smoothed time–frequency structures [15, 16, 17]. This suggests that speech generation models do not always preserve natural spectro-temporal correlations, and deviations from these correlations may manifest as artifact traces. Such artifacts may therefore provide spoofing cues complementary to attention-induced relations, yet remain underexplored in existing SDD models. Covariance neural networks (VNNs) provide a learnable approach to exploiting covariance information through polynomial filtering [18].

To address this gap, we construct a normalized affinity graph from the joint covariance of spectral and temporal representations and apply polynomial graph filtering to capture multi-hop covariance-induced dependencies. Based on this formulation, we first develop Cov-ST to examine the contribution of covariance-induced relation modeling. Although Cov-ST improves detection performance, it is sensitive to the polynomial order, indicating limited robustness when covariance relations are modeled in isolation. We therefore propose DuRe-ST, which jointly exploits covariance-induced and graph-attention-induced relations in a complementary manner. This dual-relation modeling captures both adaptive dependencies between spectral and temporal nodes and co-variation among their high-level representations. The main contributions of this work are as follows:

- We introduce covariance-induced spectro-temporal relation modeling to explicitly capture second-order dependencies between spectral and temporal representations.
- We propose Cov-ST for covariance-only relation modeling and DuRe-ST for complementary fusion with graph-attention relations.
- Experiments demonstrate that covariance-induced relations capture discriminative spoofing cues complementary to attention-induced relations, enabling DuRe-ST to achieve the best results among the compared systems across multiple benchmarks.

2. PROPOSED DURE-ST ARCHITECTURE

2.1. Shared Spectro-Temporal Nodes

Following [14], we use pretrained XLS-R 300M [8] followed by a linear projection and a residual CNN encoder to obtain $\mathbf{H} \in \mathbb{R}^{C \times F \times T}$, where C , F , and T denote the channel, frequency, and temporal dimensions. A 2-D self-attentive module produces

[†] Corresponding author: Kong Aik Lee.

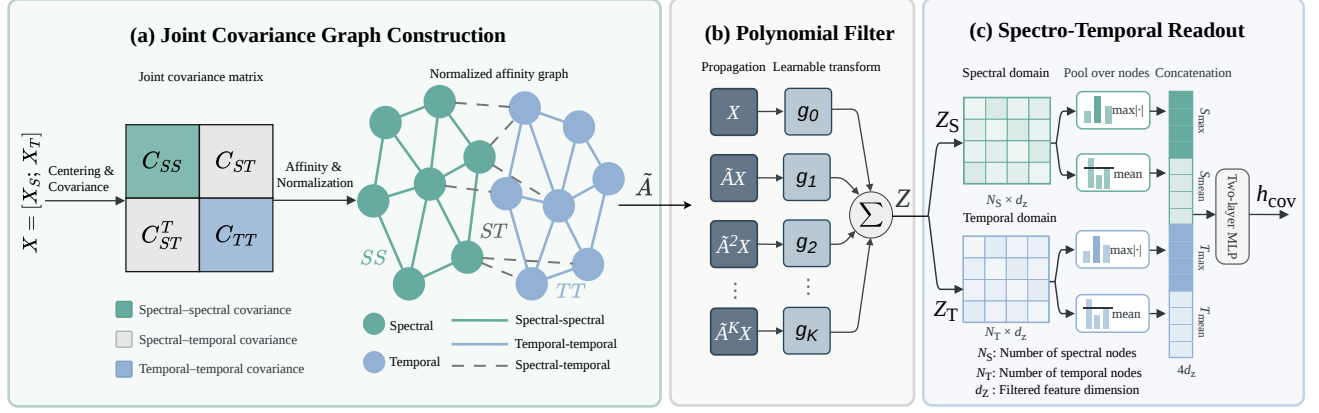


Fig. 1. Overview of covariance-induced spectro-temporal relation modeling. Given the shared node features $\mathbf{X} = [\mathbf{X}_S; \mathbf{X}_T]$, (a) constructs a normalized affinity graph $\tilde{\mathbf{A}}$ from their joint covariance, capturing second-order relations within and across the spectral and temporal domains; (b) applies an order- K polynomial graph filter over $\tilde{\mathbf{A}}$ to aggregate direct and multi-hop covariance-induced dependencies, yielding $\mathbf{Z}$; and (c) partitions $\mathbf{Z}$ into the spectral and temporal node representations $\mathbf{Z}_S$ and $\mathbf{Z}_T$, and independently summarizes each domain by max-magnitude and mean pooling over nodes. The four pooled statistics are concatenated and projected by a two-layer MLP to obtain $\mathbf{h}_{\text{cov}}$. Cov-ST uses $\mathbf{h}_{\text{cov}}$ for classification, whereas DuRe-ST concatenates it with the attention-induced representation $\mathbf{h}_{\text{att}}$ to form $\mathbf{h}_{\text{dual}}$ for classification.

logits, normalized by softmax along the temporal and frequency axes to yield $\mathbf{W}_S$ and $\mathbf{W}_T$. The corresponding representations are

$$\mathbf{U}_S = \left(\sum_t \mathbf{W}_S \odot \mathbf{H} \right)^\top, \quad \mathbf{U}_T = \left(\sum_f \mathbf{W}_T \odot \mathbf{H} \right)^\top, \quad (1)$$

where $\odot$ denotes element-wise multiplication and the sums operate along the temporal and frequency axes, respectively.

We omit AASIST’s separate spectral and temporal GAT modules and directly apply top- k graph pooling to $\mathbf{U}_S$ and $\mathbf{U}_T$, avoiding prior attention-specific transformations. Pooling reweights nodes by learned scores and retains the highest-scoring nodes, yielding compact representations $\mathbf{X}_S \in \mathbb{R}^{N_S \times C}$ and $\mathbf{X}_T \in \mathbb{R}^{N_T \times C}$. Their concatenation $\mathbf{X} = [\mathbf{X}_S; \mathbf{X}_T]$ is shared by both relation branches.

2.2. Attention-Induced Relation Modeling

For attention-induced relation modeling, we adopt the heterogeneous graph-attention module of AASIST [3, 14], which comprises a master node and two successive heterogeneous graph-attention layers. The module takes the shared spectral and temporal representations $\mathbf{X}_S$ and $\mathbf{X}_T$ as input and models intra-spectral, intra-temporal, and cross-spectro-temporal relations. Finally, the spectral and temporal representations are summarized by global maximum and average pooling and concatenated with the final master-node representation to form $\mathbf{h}_{\text{att}}$.

2.3. Covariance-Induced Relation Modeling

The covariance-induced modeling comprises joint covariance graph construction, polynomial graph filtering, and spectro-temporal readout, as illustrated in Fig. 1(a)–(c).

We treat the shared spectro-temporal nodes in $\mathbf{X}$ as variables and their $d = C$ feature dimensions as observations. After centering each node across its feature dimensions, we compute the joint covariance matrix:

$$\mathbf{C} = \frac{1}{d-1} \bar{\mathbf{X}} \bar{\mathbf{X}}^\top = \begin{bmatrix} \mathbf{C}_{SS} & \mathbf{C}_{ST} \\ \mathbf{C}_{ST}^\top & \mathbf{C}_{TT} \end{bmatrix}, \quad (2)$$

where $\bar{\mathbf{X}}$ denotes the row-wise centered representation. The diagonal blocks $\mathbf{C}_{SS}$ and $\mathbf{C}_{TT}$ capture within-domain covariance, while $\mathbf{C}_{ST}$ captures cross-domain co-variation.

To model dependency strength irrespective of sign, we use covariance magnitudes as graph affinities, add self-connections, and apply symmetric degree normalization:

$$\mathbf{A} = \text{abs}(\mathbf{C}) + \mathbf{I}_N, \quad \tilde{\mathbf{A}} = \mathbf{D}_A^{-\frac{1}{2}} \mathbf{A} \mathbf{D}_A^{-\frac{1}{2}}. \quad (3)$$

The element-wise absolute value retains the strength of both positive and negative co-variation and avoids cancellation caused by oppositely signed edge weights during aggregation. Here, $\mathbf{I}_N$ is the identity matrix, N is the total number of spectral and temporal nodes, and $\mathbf{D}_A$ is the degree matrix whose diagonal entries are the row sums of $\mathbf{A}$.

An order- K polynomial graph filter then transforms $\mathbf{X}$ over the normalized graph $\tilde{\mathbf{A}}$:

$$\mathbf{Z} = \text{SELU} \left(\text{LN} \left(\sum_{k=0}^K g_k \left(\tilde{\mathbf{A}}^k \mathbf{X} \right) \right) \right), \quad (4)$$

where $g_k(\cdot)$ denotes an order-specific linear transformation and LN denotes layer normalization. The zeroth-order term transforms each node independently, while terms with $k \geq 1$ aggregate features through covariance-induced relations over up to K propagation steps.

Finally, we split $\mathbf{Z}$ into spectral and temporal representations and summarize each domain using maximum-magnitude and mean pooling across nodes. For $q \in \{S, T\}$,

$$\mathbf{p}_q = \text{Max}(\text{abs}(\mathbf{Z}_q)) \parallel \text{Mean}(\mathbf{Z}_q), \quad (5)$$

where $\parallel$ denotes feature concatenation. The two summaries are concatenated and projected to obtain the covariance-induced representation:

$$\mathbf{h}_{\text{cov}} = f_{\text{proj}}(\mathbf{p}_S \parallel \mathbf{p}_T), \quad (6)$$

where f_{proj} is a two-layer MLP with ReLU activations and dropout.

2.4. Model Variants and Classification

Building on the relation representations derived above, we propose two models. Cov-ST uses only the covariance-induced representation, whereas DuRe-ST jointly exploits the covariance- and attention-induced representations. For DuRe-ST, $\mathbf{h}_{\text{cov}}$ and $\mathbf{h}_{\text{att}}$ are concatenated to form the dual-relation representation $\mathbf{h}_{\text{dual}}$. The corresponding representation of each model is subsequently mapped by a linear classifier to the bona fide and spoof logits.

3. EXPERIMENTAL SETUP

3.1. Datasets and Metric

All models developed in this work are trained exclusively on the ASVspoof 2019 LA training partition, which contains bona fide speech and spoofed utterances generated by six TTS/VC algorithms [19]. In this work, ASVspoof 2021 LA and DF [1] are used for the ASVspoof benchmark evaluation, whereas In-the-Wild [20], the normalized Fake-or-Real test set¹ [21], ASVspoof 5 [22], and DFADD [2] are used for the cross-dataset evaluation. All evaluations are conducted without target-domain adaptation. Performance is reported using equal error rate (EER, %), where lower values indicate better detection performance.

3.2. Implementation Details

During training, each utterance is cropped or repeatedly concatenated to approximately 4 seconds (64,600 samples). To mitigate the class imbalance between bona fide and spoofed samples, we employ weighted cross-entropy loss with class weights of 0.9 and 0.1 for bona fide and spoofed speech, respectively [3, 13]. Models are optimized using Adam with a learning rate of 10^{-6} , a weight decay of 10^{-5} , and a batch size of 16. Training stops after seven epochs without improvement in validation cross-entropy; we average the five checkpoints with the lowest validation losses for evaluation. All experiments are conducted on a single RTX 4090 GPU.

Following prior work [7, 14, 13], we train separate checkpoints using RawBoost Algo 3 and Algo 5. Algo 3 is used for ASVspoof 2021 DF and Algo 5 for the other datasets, while both are evaluated on DFADD and the lower EER is reported.

4. RESULTS AND ANALYSIS

4.1. Main Results

Table 1 compares the proposed models with the XLSR-AASIST baseline. Cov-ST models spectro-temporal dependencies using only covariance-induced relations. At $K = 2$, Cov-ST achieves a competitive EER of 1.15% on 2021 LA, compared with 1.00% for the baseline, while reducing the 2021 DF EER from 3.69% to 2.66%, a relative reduction of 27.9%. These results are obtained with fewer trainable back-end parameters than the baseline.

DuRe-ST jointly models covariance- and attention-induced relations. With $K = 1$, it achieves the best EERs of 0.93% and 2.04% on 2021 LA and DF, respectively. Relative to the baseline, DuRe-ST reduces EER by 7.0% and 44.7%, yielding an average relative reduction of 25.9% across the two benchmarks. This improvement requires only 4k additional trainable back-end parameters. Overall, the Cov-ST results demonstrate that covariance-induced relations capture discriminative spoofing cues, while the DuRe-ST results support their complementarity with attention-induced relations.

¹<https://www.kaggle.com/datasets/mohammedabdeldayem/the-fake-or-real-dataset>

Table 1. ASVspoof benchmark performance and sensitivity to the polynomial order K , reported in terms of EER (%). Params. denotes the number of trainable back-end parameters in millions. The best results are highlighted in bold.

Model	K	Params. (M)	2021 LA	2021 DF
XLSR-AASIST [14]	–	0.447	1.00	3.69
Cov-ST (Ours)	0	0.387	1.05	3.15
	1	0.391	1.56	2.71
	2	0.395	1.15	2.66
	3	0.400	1.08	7.79
DuRe-ST (Ours)	0	0.447	2.01	2.23
	1	0.451	0.93	2.04
	2	0.455	1.19	2.71
	3	0.459	2.17	2.41

4.2. Sensitivity to the Polynomial Order

The order K determines the highest power of the normalized affinity matrix included in the polynomial filter. At $K = 0$, $\tilde{\mathbf{A}}^0 = \mathbf{I}$, so the model retains only an additional transformation of the shared spectro-temporal representations without using covariance-induced information. Without covariance information, the additional transformation reduces EER from 3.69% to 2.23% on 2021 DF, but more than doubles it from 1.00% to 2.01% on 2021 LA. This substantial degradation on LA shows that the transformation alone does not yield consistent gains over the baseline. At $K = 1$, the filter incorporates covariance-induced relations, and DuRe-ST achieves its best EERs of 0.93% and 2.04% on 2021 LA and DF, respectively, outperforming both its $K = 0$ counterpart and the baseline. Extending the filter to $K = 2$ yields EERs of 1.19% and 2.71%, which remain competitive but do not improve upon $K = 1$. This comparison demonstrates that the gains stem from covariance-induced relation modeling itself, rather than merely from introducing an additional transformation of the shared spectro-temporal representations.

The benefit does not increase monotonically with K . Cov-ST achieves its best 2021 DF result at $K = 2$, but its EER increases sharply to 7.79% at $K = 3$. DuRe-ST remains comparatively stable on DF, although its 2021 LA EER also degrades to 2.17% at $K = 3$. While covariance propagation is beneficial, higher-order filtering may aggregate redundant or weak covariance relations, thereby diluting discriminative spoofing cues. We therefore retain the low-order settings $K = 1$ and $K = 2$ for cross-dataset evaluation.

4.3. Cross-Dataset Comparison with Public SDD Models

Table 2 compares Cov-ST and DuRe-ST with the XLSR-AASIST baseline and representative public SDD models. With $K = 1$, Cov-ST outperforms the baseline on all datasets except ASVspoof 5 while using 56k fewer trainable back-end parameters. It is also competitive with several public models on In-the-Wild, Fake-or-Real, and DFADD. However, these advantages diminish at $K = 2$, which offers only a modest improvement on In-the-Wild and performs worse than the baseline on the remaining datasets. At $K = 2$, the largest degradation occurs on DFADD, with a 65.8% relative increase in EER over $K = 1$. These results indicate that covariance-induced relations provide useful spoofing cues, but their effectiveness varies across datasets and filter orders. Higher-order filtering may propagate less informative relations, weakening the discriminative cues captured by covariance-only modeling.

DuRe-ST exhibits stronger and more consistent performance. With $K = 2$, it outperforms the baseline on all four datasets and achieves the best results among the compared systems on In-the-

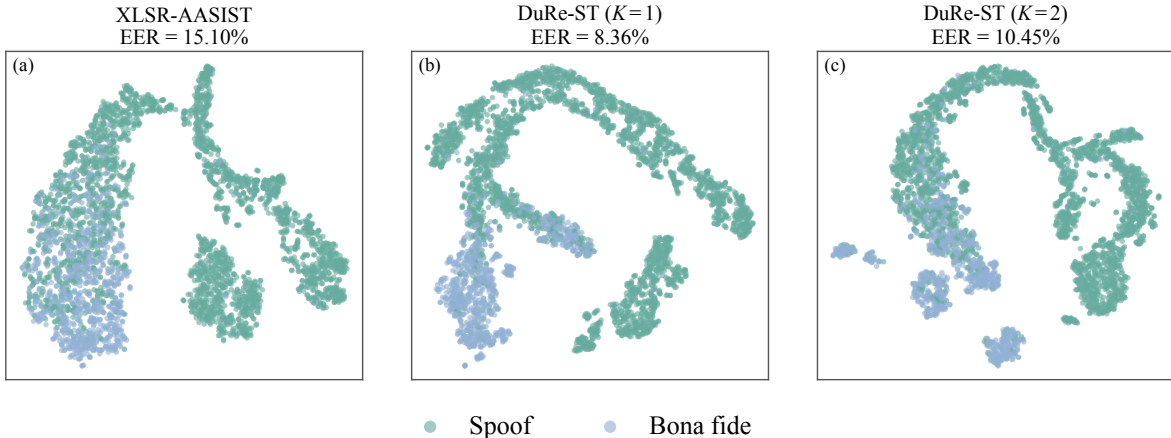


Fig. 2. t-SNE projections of utterance-level embeddings on the DFADD evaluation set for (a) XLSR-AASIST, (b) DuRe-ST with $K = 1$, and (c) DuRe-ST with $K = 2$. The corresponding EER is reported above each panel.

Table 2. Cross-dataset comparison with publicly available SDD models in terms of EER (%). $\dagger$ and $*$ denote our reproduced results and results from Speech DF Arena [23], respectively. Among the proposed models, the largest and second-largest EER reductions over the XLSR-AASIST baseline are shown in boldface and underlined, respectively.

Model	Params. (M)	In-the-Wild	Fake-or-Real	ASVspoof 5	DFADD
<i>Public models</i>					
XLSR-Nes2Net-X [11] [*]	0.512	7.75	6.31	22.05	11.14
XLSR-Mamba [13]	1.937	6.70	6.71	14.40 [*]	11.81 [†]
XLSR-Conformer [12] [†]	2.383	6.68	7.55	17.21	11.51
XLSR-Conformer-TCM [4]	2.383	7.79 [*]	10.68 [*]	18.85 [*]	10.33 [†]
XLSR-SLS [24] [†]	23.399	9.65	7.55	18.50	9.02
<i>Baseline</i>					
XLSR-AASIST [14]	0.447	11.19 [*]	7.46 [*]	16.24 [*]	15.10 [†]
<i>Ours</i>					
Cov-ST ($K = 1$)	0.391	<u>7.51</u>	<u>6.54</u>	19.94	10.47
Cov-ST ($K = 2$)	0.395	10.20	9.50	17.88	17.36
DuRe-ST ($K = 1$)	0.451	8.56	8.22	<u>16.10</u>	8.36
DuRe-ST ($K = 2$)	0.455	6.53	5.48	13.84	<u>10.45</u>

Wild, Fake-or-Real, and ASVspoof 5. Its EERs of 6.53%, 5.48%, and 13.84% improve upon the strongest public results by 2.2%, 13.2%, and 3.9%, respectively, in relative terms. On DFADD, DuRe-ST with $K = 1$ achieves the lowest EER of 8.36%, a relative reduction of 44.6% over the baseline and 7.3% over the best public result. These gains require only 4–8k additional trainable back-end parameters. With 0.451–0.455M parameters, DuRe-ST remains smaller than all public models in the comparison, whose back-ends range from 0.512M to 23.399M parameters.

Overall, the Cov-ST results highlight both the discriminative information captured by covariance-induced relations and the limitations of relying on them alone. The stronger DuRe-ST results support their complementarity with attention-induced relations for cross-dataset generalization.

4.4. Visualization of Embedding Distributions

Figure 2 visualizes the t-SNE projections of utterance-level embeddings on the DFADD evaluation set. Compared with the XLSR-AASIST baseline in Fig. 2(a), both DuRe-ST variants exhibit clearer separation between bona fide and spoofed samples. DuRe-ST with $K = 1$ in Fig. 2(b) shows the least class overlap, consistent with

its lowest EER of 8.36%. By comparison, the increased local overlap observed for $K = 2$ in Fig. 2(c) corresponds to its higher EER of 10.45%. Although t-SNE provides only qualitative evidence, the observed class separation is consistent with the EER results, suggesting that covariance-induced relation modeling contributes to a more discriminative embedding space.

5. CONCLUSION

We have introduced covariance-induced spectro-temporal relation modeling for SDD and proposed Cov-ST and DuRe-ST. Cov-ST demonstrates that second-order dependencies provide discriminative spoofing cues, although relying on these relations alone is sensitive to the polynomial order and evaluation domain. DuRe-ST achieves an average relative EER reduction of 25.9% over the baseline on the ASVspoof benchmarks and outperforms the strongest public models in the comparison by 2.2–13.2% in relative EER, with only 0.451–0.455M trainable back-end parameters. These results support the complementary role of covariance-induced spoofing cues in improving cross-dataset generalization through joint modeling with attention-induced relations. Future work will explore input-adaptive order weighting to reduce order sensitivity.

6. REFERENCES

- [1] Xuechen Liu, Xin Wang, Md Sahidullah, et al., “ASVspooF 2021: Towards spoofed and deepfake speech detection in the wild,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2507–2522, 2023.
- [2] Jiawei Du, I-Ming Lin, I-Hsiang Chiu, Xuanjun Chen, Haibin Wu, Wenze Ren, Yu Tsao, Hung yi Lee, and Jyh-Shing Roger Jang, “DFADD: The diffusion and flow-matching based audio deepfake dataset,” in *Proc. IEEE SLT Workshop*, 2024, pp. 921–927.
- [3] Jee weon Jung, Hee-Soo Heo, Hemlata Tak, Hye jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans, “AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks,” in *Proc. ICASSP*, 2022.
- [4] Duc-Tuan Truong, Ruijie Tao, Tuan Nguyen, Hieu-Thi Luong, Kong Aik Lee, and Eng Siong Chng, “Temporal-channel modeling in multi-head self-attention for synthetic speech detection,” in *Proc. Interspeech*, 2024, pp. 537–541.
- [5] Siqing Qin, Kong Aik Lee, Man-Wai Mak, Pasquale Lisena, and Massimiliano Todisco, “Variational regularization for end-to-end speech deepfake detection,” in *2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2025, pp. 2241–2246.
- [6] Duc-Tuan Truong, Tianchi Liu, Junjie Li, Ruijie Tao, Kong Aik Lee, and Eng Siong Chng, “Addressing gradient misalignment in data-augmented training for robust speech deepfake detection,” in *Proc. ICASSP*, 2026, pp. 16252–16256.
- [7] Hemlata Tak, Madhu Kamble, Jose Patino, Massimiliano Todisco, and Nicholas Evans, “RawBoost: A raw data boosting and augmentation method applied to automatic speaker verification anti-spoofing,” in *Proc. ICASSP*, 2022.
- [8] Arun Babu, Changhan Wang, Andros Tjandra, et al., “XLS-R: Self-supervised cross-lingual speech representation learning at scale,” in *Proc. Interspeech*, 2022.
- [9] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, and Furu Wei, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [10] Xu Li, Xixin Wu, Hui Lu, Xunying Liu, and Helen Meng, “Channel-wise gated res2net: Towards robust detection of synthetic speech attacks,” in *Proc. Interspeech*, 2021, pp. 4314–4318.
- [11] Tianchi Liu, Duc-Tuan Truong, Rohan Kumar Das, Kong Aik Lee, and Haizhou Li, “Nes2net: A lightweight nested architecture for foundation model driven speech anti-spoofing,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 12005–12018, 2025.
- [12] Eros Rosello, Alejandro Gomez-Alanis, Angel M. Gomez, and Antonio Peinado, “A conformer-based classifier for variable-length utterance processing in anti-spoofing,” in *Proc. Interspeech*, 2023, pp. 5281–5285.
- [13] Yang Xiao and Rohan Kumar Das, “XLSR-Mamba: A dual-column bidirectional state space model for spoofing attack detection,” *IEEE Signal Processing Letters*, vol. 32, pp. 1276–1280, 2025.
- [14] Hemlata Tak, Massimiliano Todisco, Xin Wang, Jee weon Jung, Junichi Yamagishi, and Nicholas Evans, “Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation,” in *Proc. The Speaker and Language Recognition Workshop (Odyssey)*, 2022.
- [15] Yang Gao, Tyler Vuong, Mahsa Elyasi, Gaurav Bharaj, and Rita Singh, “Generalized spoofing detection inspired from audio generation artifacts,” in *Proc. Interspeech*, 2021, pp. 4184–4188.
- [16] Xiaohui Liu, Meng Liu, Lin Zhang, Linjuan Zhang, Chang Zeng, Kai Li, Nan Li, Kong Aik Lee, Longbiao Wang, and Jianwu Dang, “Deep spectro-temporal artifacts for detecting synthesized speech,” in *Proc. the 1st International Workshop on Deepfake Detection for Audio Multimedia*, 2022, pp. 69–75.
- [17] Yi Ren, Xu Tan, Tao Qin, Zhou Zhao, and Tie-Yan Liu, “Revisiting over-smoothness in text to speech,” in *Proc. the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 8197–8213, Association for Computational Linguistics.
- [18] Saurabh Sihag, Gonzalo Mateos, Corey McMillan, and Alejandro Ribeiro, “coVariance neural networks,” in *Proc. Advances in Neural Information Processing Systems*, 2022.
- [19] Xin Wang, Junichi Yamagishi, Massimiliano Todisco, et al., “ASVspooF 2019: A large-scale public database of synthesized, converted and replayed speech,” *Computer Speech & Language*, vol. 64, pp. 101114, 2020.
- [20] Nicolas M. Müller, Pavel Czepin, Franziska Dieckmann, Adam Froggyar, and Konstantin Böttinger, “Does audio deepfake detection generalize?,” in *Proc. Interspeech*, 2022.
- [21] Ricardo Reimao and Vassilios Tzerpos, “FoR: A dataset for synthetic speech detection,” in *Proc. Speech Technology and Human-Computer Dialogue (SpeD)*, IEEE, 2019, pp. 1–10.
- [22] Xin Wang, Héctor Delgado, Hemlata Tak, et al., “ASVspooF 5: Crowdsourced speech data, deepfakes, and adversarial attacks at scale,” *arXiv preprint arXiv:2408.08739*, 2024.
- [23] Sandipana Dowerah, Atharva Kulkarni, Ajinkya Kulkarni, Hoan My Tran, Joonas Kalda, Artem Fedorchenko, Benoit Fauve, Damien Lolive, Tanel Alumäe, and Mathew Magimai-Doss, “Speech DF arena: A leaderboard for speech deepfake detection models,” *IEEE Open Journal of Signal Processing*, vol. 7, pp. 73–81, 2026.
- [24] Qishan Zhang, Shuangbing Wen, and Tao Hu, “Audio deepfake detection with self-supervised XLS-R and SLS classifier,” in *Proc. the 32nd ACM International Conference on Multimedia*, 2024, pp. 6765–6773.